RESEARCH PARTNERSHIP PROPOSAL

Moove × Mwendo Pamoja

Test how vehicle uptime changes driver cash, fee collection and the financing economics across a full contract.

Mobility finance · 156-week synthetic operating and financing case

Nevil Maloba · Research and quantitative finance · 14 October 2026

A repair outage can reduce earnings while scheduled fees continue

When a vehicle is unavailable, which operating response protects driver cash while keeping the full financing arrangement sustainable?

Observed mechanism

Fares and platform receipts depend on productive days. Repair delays can reduce available cash while contractual obligations continue.

Research question

Compare repair response and fee treatment using driver, operator, insurer and funder outcomes over the same contract horizon.

The 2023 CGAP case provides the motivation; this model turns the operating question into a reproducible test.

One repair event changes four cash ledgers

Vehicle availability → **Settled fares** → **Driver cash** → **Contract fee** → **Arrears / cure**

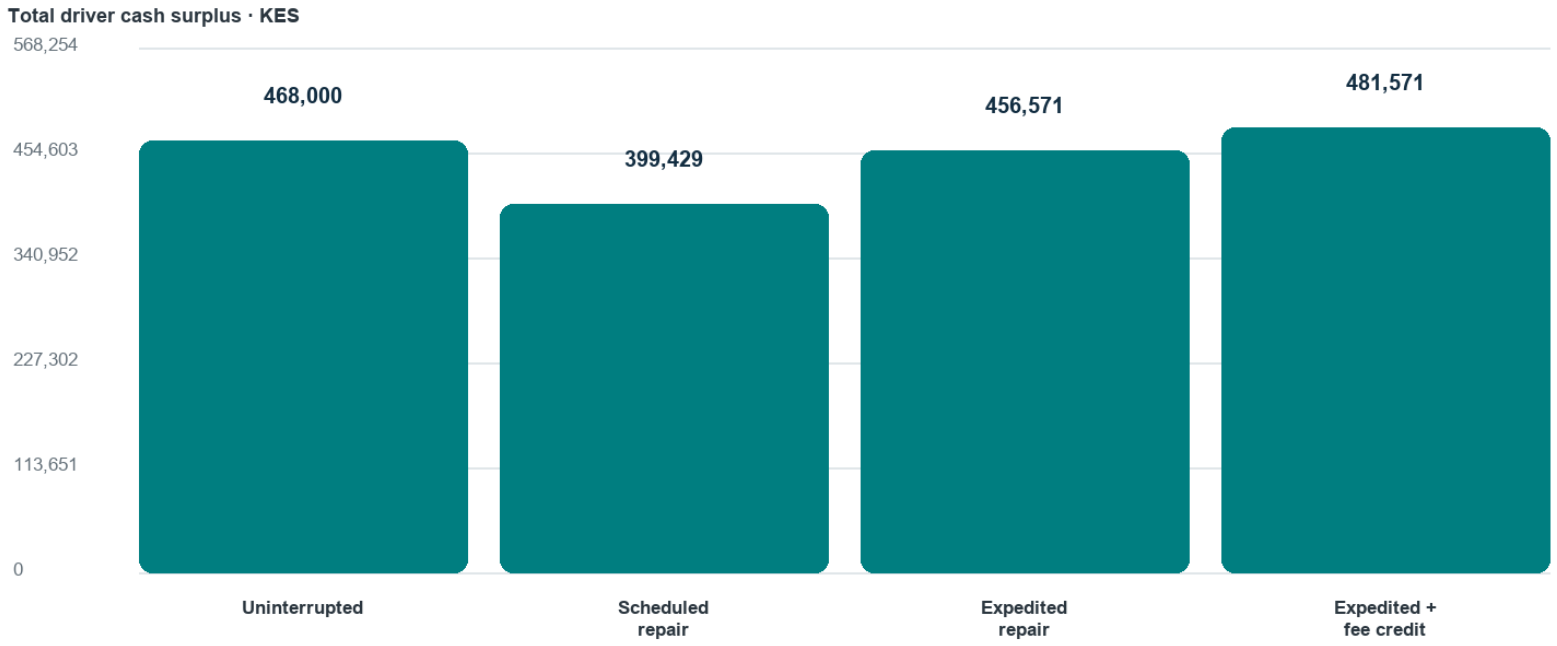
The weekly ledger reconciles the event into four views

Driver: fares, energy, household cash, fees and wallet balance
Operator: receipts, repair costs, debt service, equity calls and distributions
Funder: principal, interest, arrears and cure
Insurance: premium, claim timing, repair cost and service continuity

Scheduled repair leaves driver cash KES 68,571 below uninterrupted operation

Driver cash across a 156-week contract

Faster repair protects operating time; fee credit changes who bears the cost.



Synthetic case; author-selected assumptions. Household cash requirement met in all four paths.

The same contract fee total can sit beside different driver cash outcomes.

Scheduled repair creates KES 50,000 peak arrears

Fee timing and cure across the 156-week contract

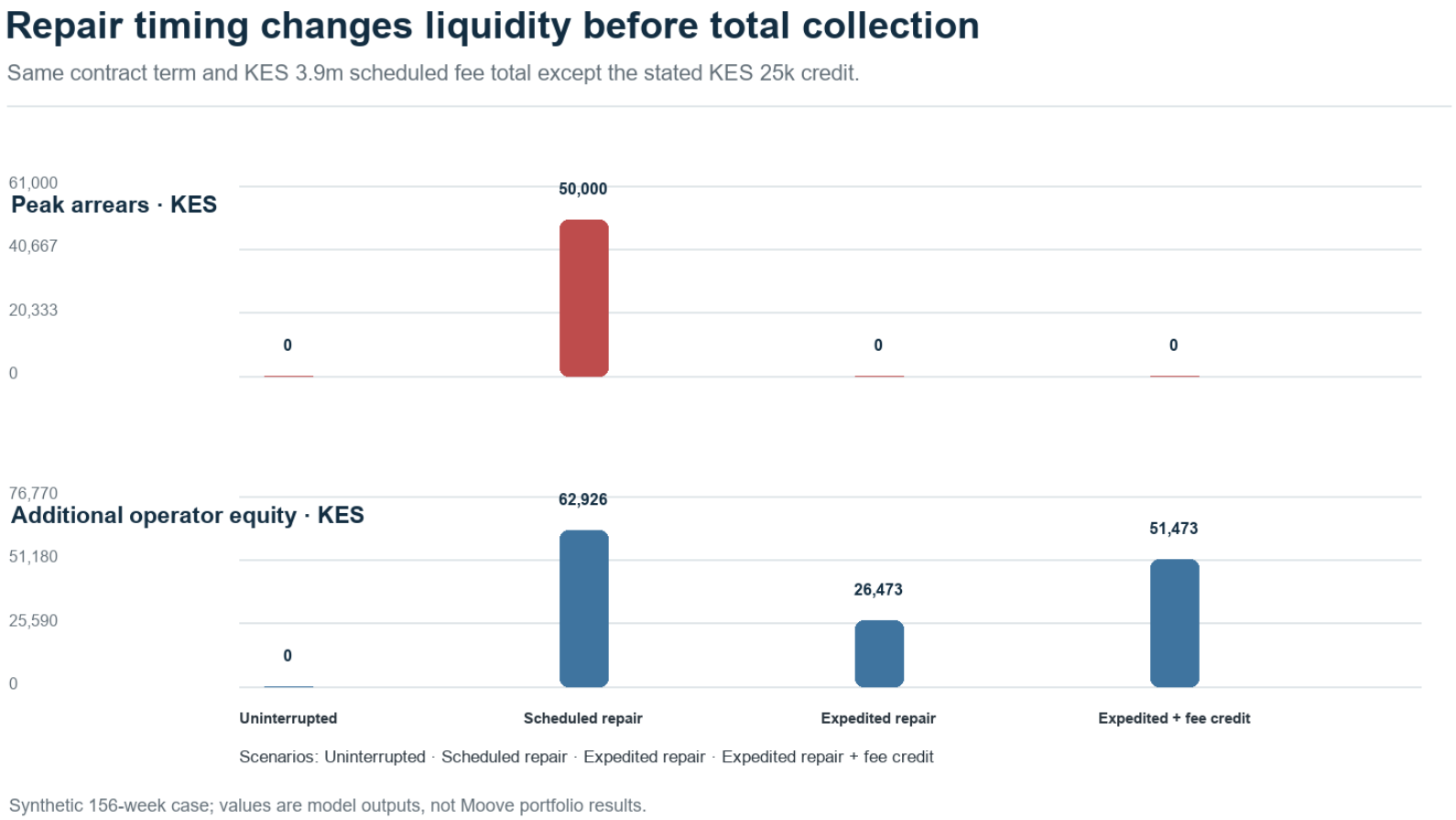
Scheduled repair creates a temporary cash gap; subsequent receipts close it.



Synthetic 156-week ledgers · KES 3.90m collected except KES 25k fee credit.

Scheduled repair reaches full collection after a KES 50,000 peak arrears balance and cure in week 32.

Expedited repair cuts the operator equity call by KES 36,453

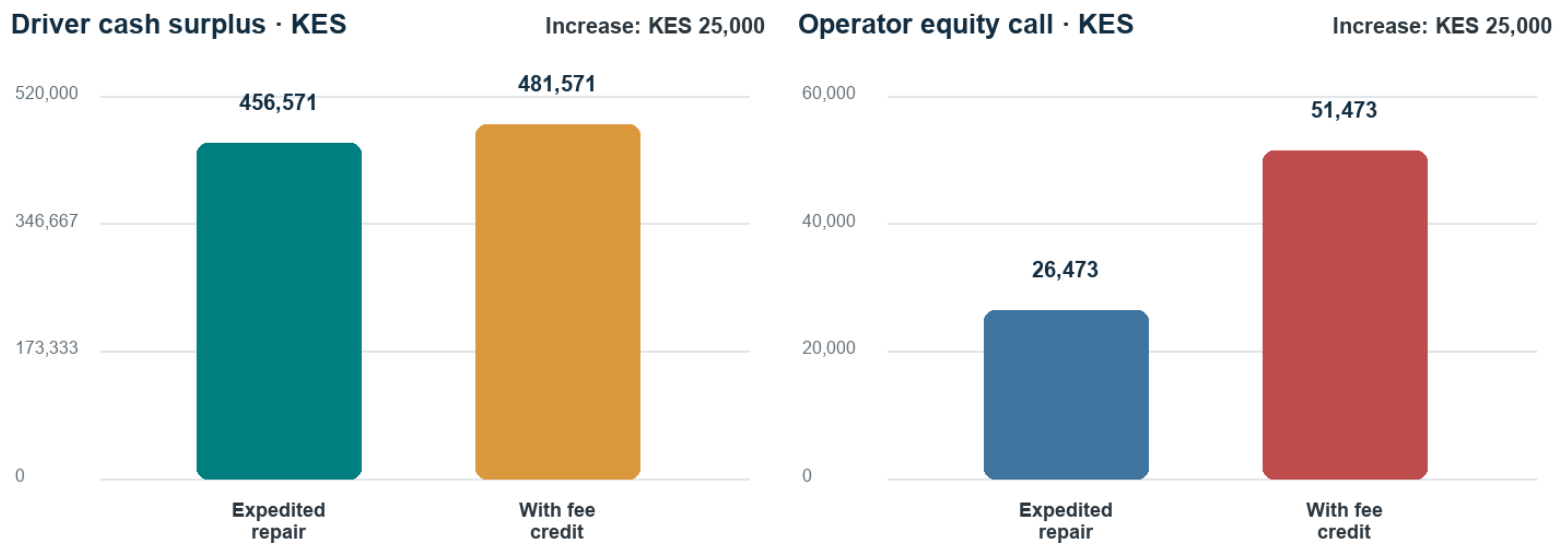


Expedited repair reduces the modeled equity call relative to scheduled repair; fee treatment changes who carries the cost.

Fee relief increases driver cash and operator capital need

A KES 25,000 fee credit moves value across two ledgers

Expedited repair compared with expedited repair plus a KES 25,000 fee credit.



The credit adds KES 25,000 to driver surplus and the same amount to the operator equity call.

Synthetic 156-week case · same vehicle and repair path; fee credit changes the allocation.

Pilot question: does the liquidity gain improve uptime and repayment enough to justify the operator cash call?

Pilot the service response that operations can deliver

Randomise eligible repair cases between the current process and one specified expedited-response offer, with safety and service rules fixed in advance.

Primary outcomes

Unavailable days; time to repair; driver net cash; on-time payment; arrears and cure.

Balancing outcomes

Operator cash and equity calls; fee waivers; claim frequency; customer complaints; service quality.

Pre-specify eligibility, assignment, follow-up, exclusions, missing-data treatment and stop conditions.

Keep participant rights and decision authority visible

Use a purpose-specific data map across platform, repair, repayment, insurance and household cash records.
Minimise identifiers; retain the linkage key under restricted access.
Record source, timestamp, event ID, model version and decision owner.
Route material changes and hardship cases to trained human review.
Separate proposed service alerts from credit-policy changes and collection authority.

The pilot needs an approved data-protection assessment, customer communication, access controls and a documented review path.

A focused design phase can produce a decision-ready pilot

First working session

Bring fleet/repair operations, credit, collections, finance, insurance, data, compliance and customer-support owners.

First deliverables

Agreed event schema; verified scenario inputs; intervention definition; outcome dictionary; pilot protocol; review gates.

Decision requested: nominate data and operating owners for a scoped feasibility workshop.

Evidence and scope

CGAP, Putting Gig Data to Work, 2023.
Mwendo Pamoja repository and 156-week mobility fleet case:
https://github.com/TheAlchemistNerd/mwendo-pamoja-continuous-underwriting
Synthetic operating and financing assumptions; weekly ledgers and verification files are retained in the research_lab and output folders.
All chart amounts are KES. Results are illustrative and depend on stated model assumptions.

Proposed collaboration question: which service and financing design can improve fleet uptime while maintaining transparent economics for each participant?